

BRANDON LAM

Student at The Ohio State University

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blam92

brandonlam2462



EXPERIENCE

Grader

Kumon

August 2022 – January 2023

- Grading classwork and homework, tutoring

Mentor

MOMENTUM Program

June 2025 – Ongoing

- Increase retention in undergraduate physics by pairing students in intermediate mechanics with upper-classmen mentors

Instructional Assistant

The Ohio State University

August 2025 – Ongoing

- Grading problem sets and midterm in astronomy course on basic astrophysics and planetary astronomy
- Grading and support for astronomy course "Life in the Universe"
- Grading and support for astronomy course "Methods of Astronomical Observation and Data Analysis"

Undergraduate Research

PHANGS

January 2024 – July 2024

- Clean up and validate galaxy images from WISE and Spitzer
- Compare to JWST images to ensure background level is correct

Undergraduate Research

DESI

May 2025 – July 2025

- Take simulation data of the intergalactic medium and forward-model it to look like DESI data
- Analyze optical depth distributions and see how much data would be needed to constrain helium reionization

Undergraduate Research

Course-Based Undergraduate Research

October 2025 – December 2025

- Query Gaia data of the IC4665 open cluster
- Examine how different membership selection affects the interpretation of the cluster's structure

Undergraduate Research

DESI

February 2026 – May 2026

- Use data from the PRIYA suite of cosmological simulations
- Investigate potential use of machine learning to estimate the mean temperature of the intergalactic medium

Undergraduate Research

NRAO

May 2026 – August 2026

- Anchor newly observed strip of JWST data to WISE
- Compare new VLA observations to old VLA data from THINGS
- Use new observations and archival JWST, ALMA, GALEX, Spitzer, and WISE data to investigate neutral atomic and molecular gas, star formation, and polycyclic aromatic hydrocarbons in NGC 628

PROGRAMMING LANGUAGES

Python



Java



R



EDUCATION

High School Diploma

William Mason High School

August 2019 – May 2023

Bachelor of Science

The Ohio State University

August 2023 – In Progress

RELEVANT COURSEWORK

Math

- AP Calculus AB
- Math 1152: Calculus 2
- Math 2153: Calculus 3
- Math 2415: Ordinary and Partial Differential Equations
- Math 2568: Linear Algebra

Physics

- Physics 1270 and 1271: Introductory Physics
- Physics 2300 and 2301: Intermediate Mechanics and Special Relativity
- Physics 3700: Data Analysis Lab
- Physics 5500H: Honors Quantum Mechanics I
- Physics 5680: Big Data and Machine Learning
- Physics 5400: Electricity and Magnetism I
- Physics 5600: Statistical Mechanics
- Physics 5700: Advanced Physics Lab (ongoing)
- Physics 3470: Optics (planned)
- Physics 5810: Computational Physics (planned)

Astronomy

- Astronomy 1221: Astronomy Data Analysis
- Astronomy 2291: Basic Astrophysics and Planetary Astronomy
- Astronomy 2292: Stellar, Galactic, and Extragalactic Astronomy and Astrophysics
- Astronomy 3350: Methods of Astronomical Observation and Data Analysis
- Astronomy 5681: Principles of Stellar Evolution and Nucleosynthesis
- Astronomy 5862: Cosmology (ongoing)
- Astronomy 5550: Advanced Astronomical Data Analysis (planned)

Computer Science

- CSE 1223: Introduction to Computer Programming in Java
- CSE 2123: Data Structures Using Java
- CSE 2321: Foundations I–Discrete Structures
- CSE 4471: Information Security
- CSE 3430: Overview of Computer Systems For Non-Majors (ongoing)
- CSE 3241: Introduction to Database Systems (planned)